

Predicting Marathon Race Performance Using Machine Learning



Sanad Sahawneh, Brian Allen, and Mahib Rahman

ECE465: Project Proposal Presentation

Faculty Mentor: Dr. Hany Elgala

MOTIVATION

- Marathon performance depends on multiple factors such as age range, gender, and pacing strategy.
- Predicting finish times can help runners set realistic goals and optimize training.
- Real race data provides a practical way to apply machine learning to a familiar, real-world problem.
- Stakeholders: runners, coaches, and sports analysts who want performance predictions.

PROJECT GOAL

- Predict a runner's net marathon finish time.
- Use demographic and race-split features from the 2023 Boston Marathon dataset of 26,598 runners.
- Compare a classical model and a neural network to evaluate prediction accuracy.

SYSTEM OVERVIEW

- Input: Runner features (age range, gender, etc.).
- Preprocessing: Handle missing values and encode categorical variables.
- Models: linear regression (baseline) and neural network regression
- Output: Predicted finish time (seconds).

METHODOLOGY

- Clean and preprocess the dataset.
- Split data into training and testing sets.
- Train a linear regression baseline model.
- Train a neural network regression model.
- Compare performance using evaluation metrics.

EVALUATION & OUTCOMES

- Metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE)
- Compare neural network performance to linear regression baseline.
- Expected outcome: Neural network achieves lower prediction error by capturing non-linear relationships.

Names: [Sanad Sahawneh](#), [Brian Allen](#), and [Mahib Rahman](#)

IECE 465 Project Proposal: Machine Learning-Based Prediction of Marathon Race Performance

Faculty Mentor: Dr. Hany Elgala

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Introduction: Predicting athletic performance is an important problem in sports analytics and training. For endurance events like marathons, finish time depends on both athlete characteristics and in-race performance. We chose this project because it combines machine learning with a real, familiar athletic application and allows us to work with realistic race data rather than idealized examples. In this project, we are using machine learning to predict marathon finish times using race results from the [2023 Boston Marathon](#).

Problem Specification: The goal of this project is to predict a runner's net marathon finish time using demographic and performance-related features such as age range, gender, and half-marathon split time. This is formulated as a supervised regression problem, where the output is the runner's finish time in seconds.

Data: We are using the Boston Marathon 2023 dataset, which contains race results for thousands of runners. The dataset includes demographic information, intermediate split times, and final finish times. Because the data comes from a real race, it captures natural variability in athletic performance and pacing. The target variable for prediction is *finish_net_sec*.

Approach: We will clean and preprocess the data by handling missing values and encoding categorical features. We will start with a classical linear regression model as a baseline, which assumes a simple relationship between runner features and finish time. We will then implement a neural network regression model to learn more complex, non-linear patterns in the data. Comparing these models will show how increased complexity affects prediction accuracy and whether the neural network improves performance over the baseline.

Evaluation Criteria: Model performance will be evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics provide a clear way to compare models and measure improvements over the baseline.

Expected Conclusions: We expect more flexible models to outperform simpler linear models by capturing nonlinear relationships in the data. This project aims to show how machine learning can be effectively applied to real-world athletic performance prediction.

References:

1. SCORE Sports Data Repository – [2023 Boston Marathon runners](#). Provides the full dataset of 26,598 runners, including age group, gender, placements, and half/full marathon times, used for all calculations, graphs, and statistical analysis in the project.